

Investment Monte Carlo Projection - Data Flow Documentation

This document maps the complete data flow for Monte Carlo projections, using the Mitchell persona (peak_earners) as a working example.

Mitchell Portfolio Overview

Investment Accounts

Account	Owner	Type	Value	Risk Preference	Custom Risk
David's S&S ISA	David	ISA	£95,000	high	Yes
Sarah's S&S ISA	Sarah	ISA	£85,000	medium	No
Joint GIA	Joint	GIA	£95,000	upper_medium	No
VCT Holdings	David	VCT	£30,000	high	Yes

DC Pensions (also use Monte Carlo)

Account	Owner	Type	Value	Risk Preference
Global Finance Corp Pension	David	Workplace	£180,000	upper_medium
David's SIPP	David	SIPP	£320,000	upper_medium

User Risk Profiles

User	Main Risk Level	Source
David	upper_medium	Self-assessed
Sarah	medium	Self-assessed

1. Portfolio-Level Monte Carlo

Data Sources

File: `app/Services/Investment/InvestmentProjectionService.php` **Method:** `getPortfolioProjections()` (lines 51-94)

Step 1: Calculate Total Portfolio Value

Portfolio Value = $\sum (\text{Account Value} \times \text{Ownership Percentage} / 100)$

For Mitchells (David's view):

David's S&S ISA:	£95,000 × 100% = £95,000
Sarah's S&S ISA:	£85,000 × 0% = £0 (spouse-owned)
Joint GIA:	£95,000 × 50% = £47,500
VCT Holdings:	£30,000 × 100% = £30,000
Total Portfolio Value:	£172,500

Source: `InvestmentAccount` model fields:

- `current_value` (float)
- `ownership_type` (individual/joint/trust)
- `ownership_percentage` (decimal)

Step 2: Calculate Weighted Average Risk

File: `app/Services/Investment/InvestmentProjectionService.php` **Method:** `calculateWeightedPortfolioRisk()` (lines 322-368)

Risk Levels → Numeric Values:

low	= 1
lower_medium	= 2
medium	= 3
upper_medium	= 4
high	= 5

For Mitchells (David's view):

Account	Value	Weight	Risk	Numeric	Weighted
David's S&S ISA	£95,000	55.1%	high	5	2.755
Joint GIA (50%)	£47,500	27.5%	upper_med	4	1.100
VCT Holdings	£30,000	17.4%	high	5	0.870
Total	£172,500	100%			4.725

Weighted Risk = 4.725 → Rounds to "upper_medium" (4–5 range)

Source: Each account's risk determined by:

1. `risk_preference` field (if `has_custom_risk = true`)
2. User's main `risk_level` from `RiskProfile` model
3. Default: "medium"

Step 3: Get Return Parameters for Portfolio Risk

File: `app/Services/Risk/RiskPreferenceService.php` **Method:** `getReturnParameters()` (lines 314-324)

Risk Level	Expected Return (Typical)	Volatility
low	2.0%	3%
lower_medium	3.5%	6%
medium	5.0%	10%
upper_medium	6.5%	15%
high	8.0%	20%

For Mitchells' portfolio (upper_medium):

- **Expected Return:** 6.5%
- **Volatility:** 15%

Step 4: Estimate Portfolio Contributions

File: `app/Services/Investment/ContributionEstimatorService.php`

Account Type	Estimation Method
ISA	<code>isa_subscription_current_year ÷ months_elapsed</code> OR <code>£20,000 ÷ 12</code>
GIA	<code>current_value × 5% ÷ 12</code>
VCT	£0 (typically no regular contributions)

For Mitchells:

David's S&S ISA:	£10,000 ÷ 10 months = £1,000/month
Joint GIA (50%):	£47,500 × 5% ÷ 12 = £198/month
VCT Holdings:	£0/month

Estimated Monthly: £1,198/month

Step 5: Run Monte Carlo Simulation

File: `app/Services/Investment/MonteCarloSimulator.php` **Method:** `simulate()` (lines 20-94)

Input Parameters:

```
$simulator->simulate(
    startValue: 172500,           // Portfolio value
    monthlyContribution: 1198,    // Estimated contributions
    expectedReturn: 0.065,        // 6.5% annual
    volatility: 0.15,             // 15% annual
    years: 20,                   // Projection period
    iterations: 1000             // Simulation runs
);
```

Algorithm:

1. Convert annual → monthly: `monthlyReturn = 0.065/12`, `monthlyVol = 0.15/√12`
2. For each of 1,000 iterations:
 - For each month:
 - Generate random return using Box-Muller normal distribution
 - `portfolio = portfolio × (1 + randomReturn) + monthlyContribution`
 - Store year-end values
3. Calculate percentiles across all iterations

Output: Year-by-year percentile bands (p5, p10, p15, p20, p50, p75, p90)

2. Individual Account Monte Carlo

Data Sources per Account

File: `app/Services/Investment/InvestmentProjectionService.php` **Method:** `calculateAccountProjection()` (lines 156-226)

Risk Priority Hierarchy

1. Account has `risk_preference` AND `has_custom_risk = true`
→ Use account's `risk_preference`
2. Account has `risk_preference` but `has_custom_risk = false`
→ Use user's main `risk_level`
3. Account has no `risk_preference`
→ Use user's main `risk_level`
4. User has no risk profile
→ Default to "medium"

Mitchell Individual Account Projections

David's S&S ISA:

```
Value:           £95,000 × 100% = £95,000
Risk Source:     Account-level (has_custom_risk = true)
Risk Level:      high
Expected Return: 8.0%
Volatility:      20%
Est. Contribution: £1,000/month (from ISA subscription)
```

Sarah's S&S ISA (spouse account, not in David's portfolio):

Value:	$\pounds 85,000 \times 100\% = \pounds 85,000$
Risk Source:	Account-level
Risk Level:	medium
Expected Return:	5.0%
Volatility:	10%
Est. Contribution:	$\pounds 833/\text{month} (\pounds 10,000 \div 12)$

Joint GIA (David's 50% share):

Value:	$\pounds 95,000 \times 50\% = \pounds 47,500$
Risk Source:	Account-level
Risk Level:	upper_medium
Expected Return:	6.5%
Volatility:	15%
Est. Contribution:	$\pounds 198/\text{month} (5\% \text{ of share } \div 12)$

VCT Holdings:

Value:	$\pounds 30,000 \times 100\% = \pounds 30,000$
Risk Source:	Account-level (has_custom_risk = true)
Risk Level:	high
Expected Return:	8.0%
Volatility:	20%
Est. Contribution:	$\pounds 0/\text{month}$

Monte Carlo per Account

Each account runs independent simulation with its own parameters:

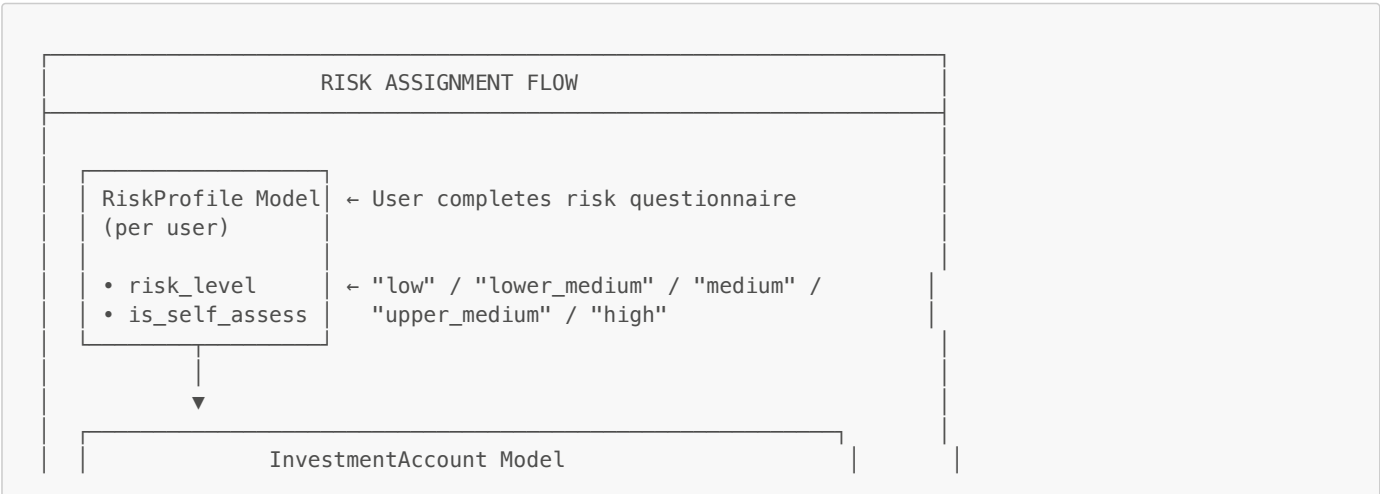
```
// David's S&S ISA
$simulator->simulate(95000, 1000, 0.08, 0.20, 20, 1000);

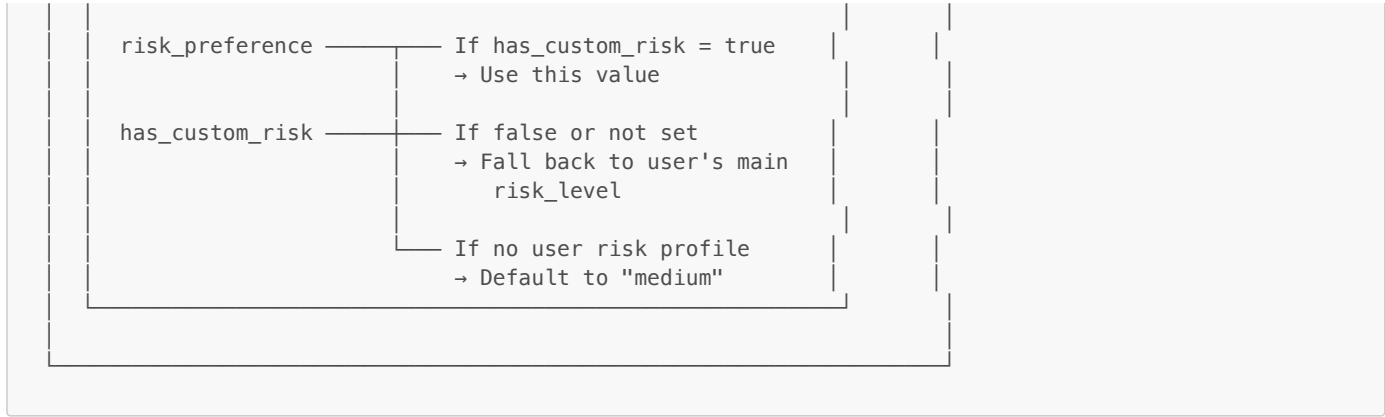
// Joint GIA (David's share)
$simulator->simulate(47500, 198, 0.065, 0.15, 20, 1000);

// VCT Holdings
$simulator->simulate(30000, 0, 0.08, 0.20, 20, 1000);
```

3. Risk Assignment Flow

How Risk Gets to Each Account





Risk Constraints

File: `app/Services/Risk/RiskPreferenceService.php` Method: `getAllowedProductRiskLevels()` (lines 216-236)

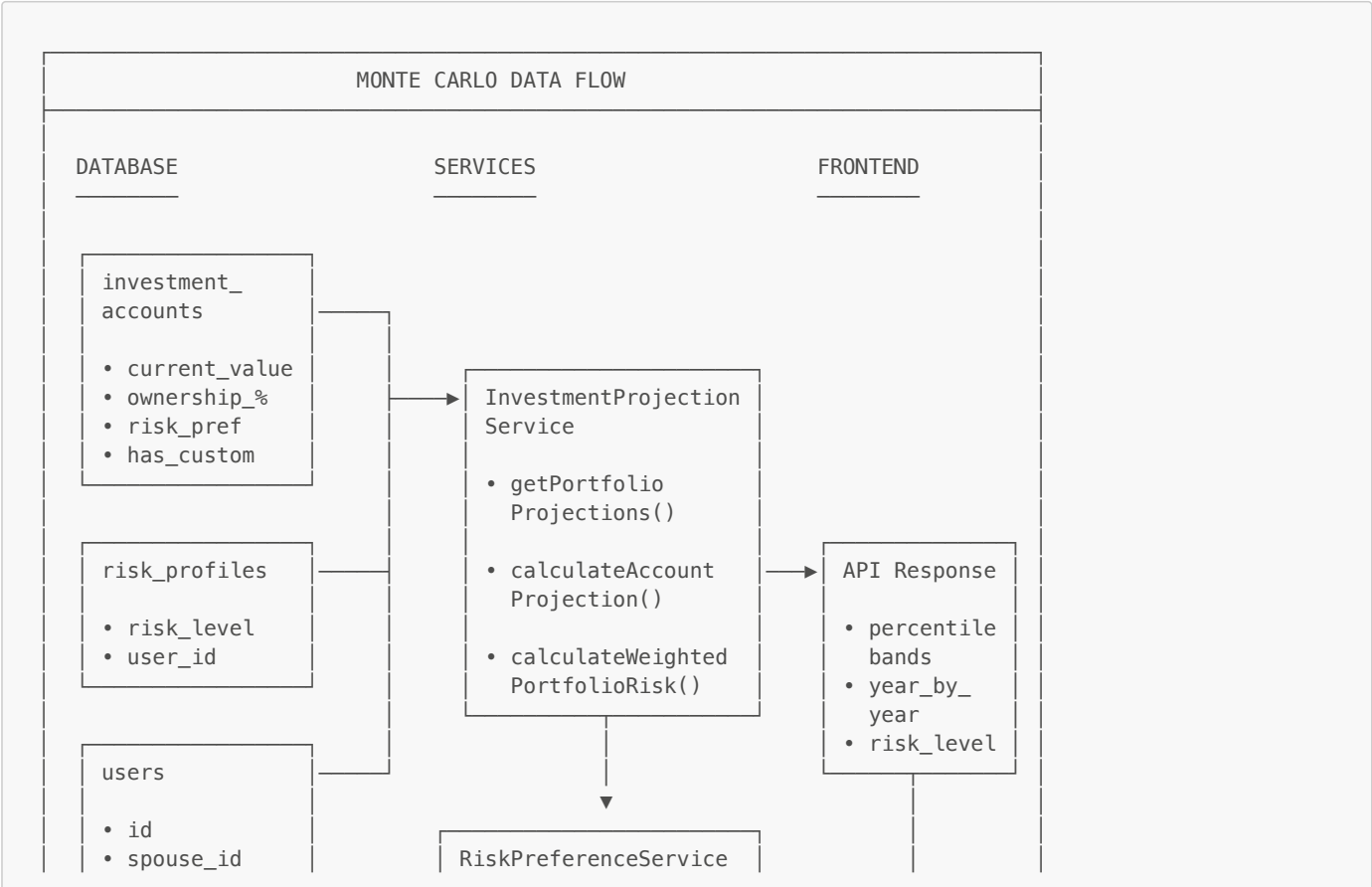
Account-level risk must be within ±1 of user's main risk level:

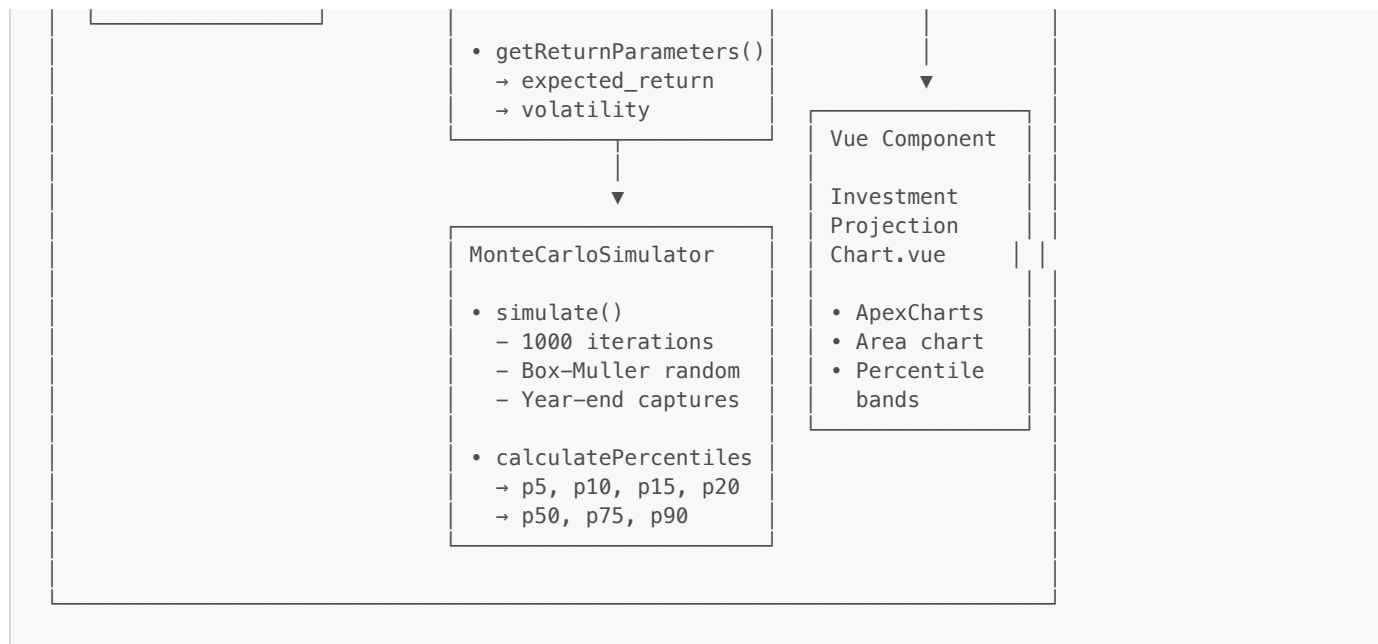
User Main Risk	Allowed Account Risks
low	low, lower_medium
lower_medium	low, lower_medium, medium
medium	lower_medium, medium, upper_medium
upper_medium	medium, upper_medium, high
high	upper_medium, high

For David (main: upper_medium):

- Can set accounts to: medium, upper_medium, or high ✓
- His "high" ISA is valid (within ±1)

4. Data Flow Diagram





5. API Endpoints

Portfolio Projections

POST /api/investment/projections

Request:

```
{
  "projection_periods": [5, 10, 20, 30],
  "selected_period": 20,
  "contribution_overrides": {
    "1": 1200, // Override account ID 1 contribution
    "3": 500  // Override account ID 3 contribution
  }
}
```

Response:

```
{
  "portfolio": {
    "total_value": 172500,
    "risk_level": "upper_medium",
    "expected_return": 6.5,
    "volatility": 15,
    "projections": {
      "20": {
        "years": 20,
        "median_value": 892450,
        "percentiles": {...},
        "year_by_year": [...]
      }
    }
  },
  "accounts": [
    {
      "account_id": 1,
      "account_name": "David's S&S ISA",
      "risk_level": "high",
      "projections": {...}
    },
    ...
  ]
}
```

```
GET /api/investment/accounts/{id}/projections?risk_level=medium

Response:
{
  "account_id": 1,
  "account_name": "David's S&S ISA",
  "risk_level": "medium",           // Override applied
  "risk_source": "override",
  "expected_return": 5.0,
  "volatility": 10,
  "projections": {...}
}
```

6. Key Files Reference

Component	File	Key Methods
Monte Carlo Engine	app/Services/Investment/MonteCarloSimulator.php	simulate(), calculatePercentiles()
Portfolio Projections	app/Services/Investment/InvestmentProjectionService.php	getPortfolioProjections(), calculateAccountProjection()
Risk Parameters	app/Services/Risk/RiskPreferenceService.php	getReturnParameters(), getMainRiskLevel()
Contribution Estimation	app/Services/Investment/ContributionEstimatorService.php	estimateAccountContribution()
Controller	app/Http/Controllers/Api/InvestmentController.php	getAccountProjections(), startMonteCarlo()
Frontend Chart	resources/js/components/Investment/InvestmentProjectionChart.vue	ApexCharts area chart
Results Display	resources/js/components/Investment/MonteCarloResults.vue	Full results modal

7. Risk Level Parameters Summary

Risk Level	Numeric	Expected Return	Volatility	Equity %	Bond %
low	1	2.0%	3%	10%	70%
lower_medium	2	3.5%	6%	30%	55%
medium	3	5.0%	10%	50%	40%
upper_medium	4	6.5%	15%	75%	20%
high	5	8.0%	20%	90%	5%

8. Mitchell Example: 20-Year Portfolio Projection

Input:

- Start Value: £172,500
- Monthly Contribution: £1,198
- Expected Return: 6.5%
- Volatility: 15%
- Years: 20
- Iterations: 1,000

Output (example percentiles at year 20):

- 10th percentile: £485,000 (poor market scenario)

- 25th percentile: £620,000
- 50th percentile (median): £892,000
- 75th percentile: £1,280,000
- 90th percentile: £1,850,000 (strong market scenario)

Total contributions over 20 years: $£172,500 + (£1,198 \times 240) = £460,020$ Median growth: $£892,000 - £460,020 = £431,980$ (growth from returns)